MİİS Macroeconomic Forecasting, Budget Linkage and Advanced Analytics

Final Technical Report and Implementation Package

Prepared for the Ministry of Economy MİİS workstream

31 May 2026

Decision message. The technically defensible delivery is a core-first MİİS forecasting system: governed data marts, statutory macro outputs, GDP and trade pilot models, scenario engine, model registry, diagnostics, and an explicit path to budget and advanced analytics modules. The package is ready for official discussion. It does not claim production budget execution or micro-level AI before the required internal administrative data are supplied.

Forecast vintage rule. The available official 2025 actual overlay is used where present. The first forecast year is therefore 2026. This convention must be applied consistently in the MİİS screens, reports and model runs.

Evidence rule. No synthetic data result is reported as an empirical Ministry finding. Experimental or illustrative models are labelled as such.

Contents

1 Executive Summary

The supplied e-mails, model documents, CAEM workbook, methodology files and legal references define a system implementation task rather than a single econometric exercise. The Ministry requires a MİİS capability that can support medium-term macroeconomic forecasting, scenario analysis, statutory budget preparation and advanced analytics. The immediate response expected by 1 June is to determine the technical solution, identify questions and data needs, and propose a feasible implementation path.

The recommended delivery is not to implement every listed method at once. The technically sound sequence is:

1. establish source governance, data marts and model registry;
2. implement GDP and trade as the first pilot blocks because these were explicitly prioritised in the supplied e-mails;
3. reproduce statutory macro outputs and CAEM-style consistency checks;
4. add scenario analysis and recursive VAR impulse-response tools with transparent identification;
5. link to budget modules after tax, customs, treasury, debt and fund data are supplied;
6. gate AI, deep learning, VECM, CGE, DSGE and input-output modules by data availability and out-of-sample performance.

The current evidence base is substantial. The controlled data package includes 54 CAEM sheets, 100 quarterly observations from 2001Q1 to 2025Q4, 5 reduced-form regression models, 92 quarterly observations for the recursive VAR, and 17 executed notebooks. The 2025 official actual overlay records real GDP growth of 1.40%, non-oil GDP growth of 2.70%, and CPI inflation of 5.60%. In the assembled forecast panel, the average 2026–2030 growth path is 2.44% for real GDP and 3.47% for non-oil GDP.

The final recommendation is to approve the GDP and trade pilot, define data owners for each data mart, and request the internal datasets required for the production budget engine. The full production system is feasible only if the Ministry provides governed administrative data for tax, customs, treasury execution, debt, SOFAZ/off-budget funds and project-level public investment.

2 Mandate and Interpretation of the Request

The project material combines four sources of obligation:

- the legal budget-preparation framework under Cabinet of Ministers Decree No. 75 and related budget rules;
- the Ministry's existing macroeconomic methodology and Excel/EViews-style model workbooks;
- the CAEM financial-programming workbook and documentation;
- the Databiz model notes for GDP, trade, FDI/capital flows and labour-market analytics.

The e-mail instructions clarify the operational scope. The client asks the team to review the models and methodology, work with Databiz/Ruslan müəllim, define the technical solution, identify questions or missing information, and respond by 01.06.2026. The e-mails also state that concrete models were not fixed in the technical specification; model choice must therefore be agreed with the client. This gives the implementation team responsibility to choose a method sequence that is feasible, evidence-based and consistent with the legal framework.

Table 1: Interpretation of supplied e-mail and document requests

ID	Client request	Required response	Status
EMAIL-01	Review supplied models and methodology and provide questions/additional data needs by 01.06.2026	1 June technical response memo, question list, data request, implementation priorities	in scope
EMAIL-02	Determine technical solution with Ruslan/Databiz for advanced analytics models	MİİS solution architecture, model engine choice, data mart mapping, screens/workflows	in scope
EMAIL-03	GDP and foreign trade models should be worked into MİİS quickly under implementation plan	Prioritise GDP and trade as pilot model blocks	highest priority
EMAIL-04	Budget formation depends on macroeconomic and microeconomic forecasting module	Macro-to-budget interface and data requirements for budget module	in scope with data caveat
EMAIL-05	System suggested in the example of EViews	Functional equivalent: data workfiles, equations, estimation, diagnostics, scenarios, forecast tables, audit trail	interpret as functional benchmark
DOC-01	Forecast legally required macro indicators and support budget process timetable	legal indicator matrix and statutory output generator design	in scope
DOC-02	Use CAEM-style financial programming consistency and scenario simulation	CAEM integration layer and scenario engine	in scope
DOC-03	Consider ARIMA, VAR/SVAR/VECM, Bayesian, IO/CGE/DSGE, AI/ML, risk intelligence, dashboards	model governance: promote only evidence-supported methods; keep advanced methods gated	in scope, evidence-gated

3 Source and Data Foundation

The available evidence is adequate for a serious macro-forecasting and scenario prototype. It is not adequate, by itself, for a full production budget-execution system. The distinction matters because a public aggregate fiscal series can support an illustrative budget path, but cannot support tax-policy microsimulation, customs revenue modelling, sequestration monitoring or debt-registry risk analysis.

3.1 Available Data

The controlled source inventory covers Ministry workbooks, CAEM, statutory templates, public IMF/World Bank/NSDP-style datasets, prior analytical outputs and notebook data. The strongest available blocks are:

- annual macro panel with GDP, oil/non-oil GDP, population, oil price and inflation;
- quarterly macro-financial panel covering 2001Q1–2025Q4;
- CAEM workbook inventory and extracted variable catalogue;
- regression, scenario, VAR, VECM, ML and backtest result files from prior execution;
- Databiz 13-model requirement matrix and MİİS functional mapping.

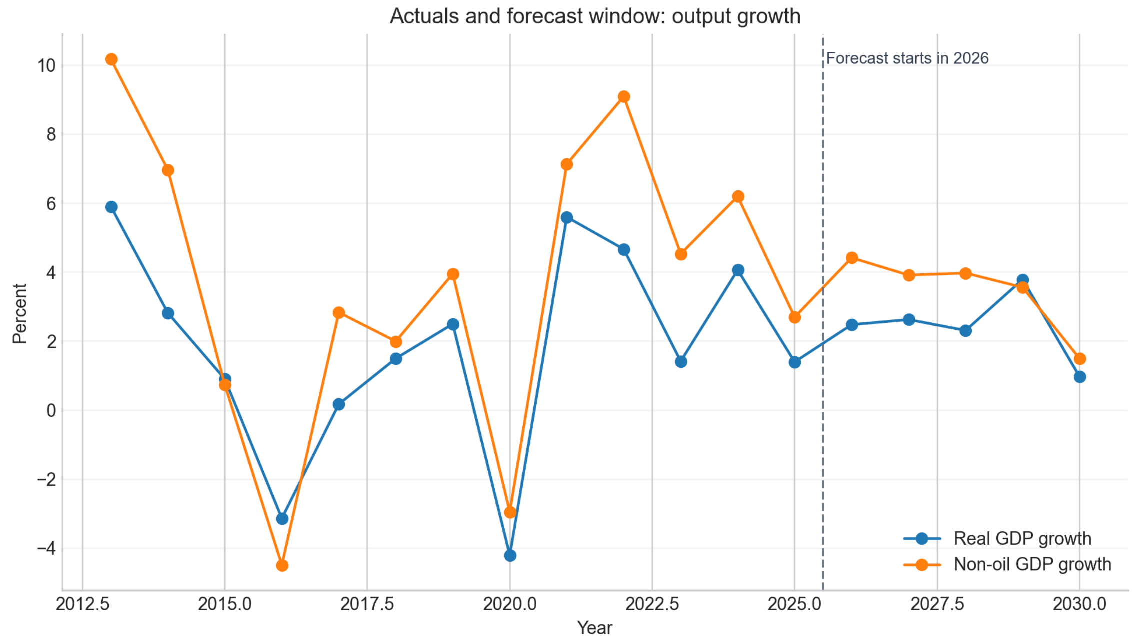

Figure 1: Actuals and forecast window for output growth. The vertical line marks the transition: 2025 is treated as actual where official actuals are available; 2026 is the first forecast year.

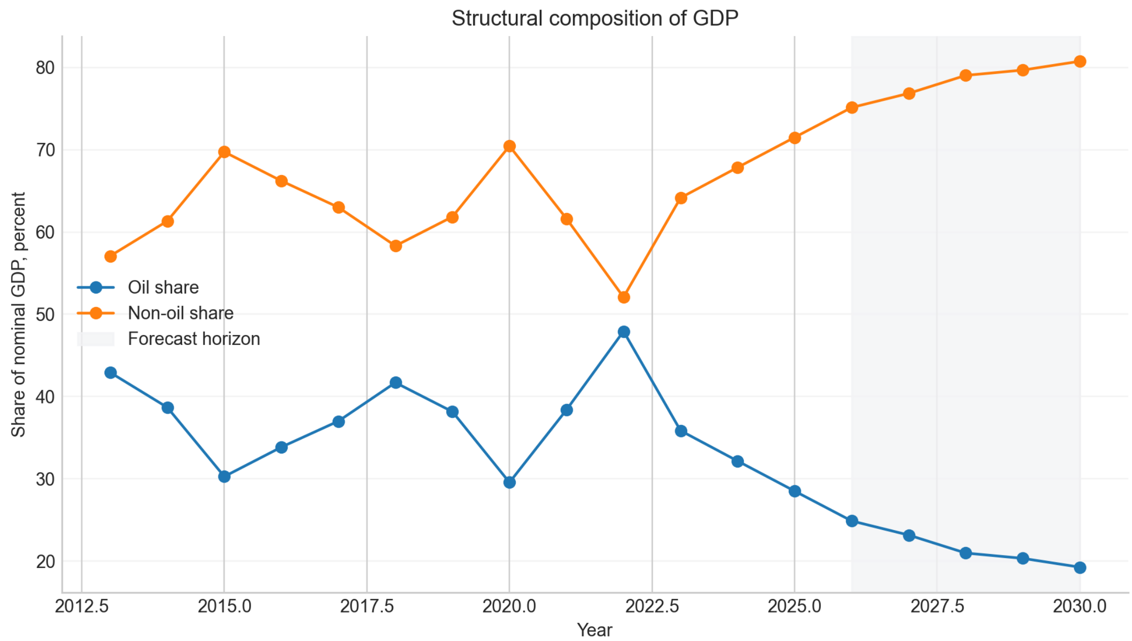

Figure 2: Oil and non-oil shares of nominal GDP in the assembled macro panel. The path supports the prioritisation of non-oil GDP modelling in the pilot.

3.2 Production Data Still Required

For the production system, the central missing component is not another public macro series. It is internal administrative data: tax assessments and refunds, customs records, treasury execution, debt registry, fund-level data and project-level investment plans. These are not optional because they define the legal and operational budget module.

Table 2: Production data required beyond public aggregate data

ID	Module	Required series	Reason	Current workaround
GAP-01	Budget revenue	Taxpayer-level or tax-type revenue assessment, collection, refund, arrears, exemption and sector-region breakdowns	Needed for production revenue forecasts, tax-policy simulations, compliance risk and legal budget module; public aggr...	Use public aggregate annual/monthly tax revenue only for prototype.
GAP-02	Customs / trade	HS/product/partner/month import-export values, duties, VAT at customs, exemptions and tariff measures	Needed for Databiz trade models, import/export decomposition, customs revenue and shock transmission by product/partner.	Use aggregate trade/BOP/customs receipt series.
GAP-03	Treasury execution	Monthly/quarterly expenditure execution by economic, functional and administrative classification; treasury cash bala...	Needed for budget execution, sequestration monitoring, and Decree-75 reporting formats.	Use public aggregate budget execution where available.
GAP-04	Debt and financing	Debt registry by instrument, currency, maturity, interest, creditor, amortisation and new borrowing plan	Needed for deficit financing, debt dynamics, refinancing risk and stress tests.	Use IMF/WEO/public debt ratios for prototype debt path.
GAP-05	SOFAZ and off-budget funds	Transfers, assets, investment income, fund inflows/outflows, off-budget entities and pension/social fund detail	Needed for consolidated budget and oil-revenue/fiscal-rule analysis.	Use aggregate fiscal/oil revenue proxies.
GAP-06	Public investment	Project-level capital expenditure pipeline, import content, implementation schedule, financing source and sector clas...	Needed for GDP/investment multipliers, import demand and medium-term fiscal planning.	Use aggregate investment and capital expenditure.
GAP-07	FDI/capital flows	FDI by sector, origin country, project, reinvested earnings, debt/equity split; other investment and portfolio detail	Needed for Databiz FDI/capital-flow models and policy interpretation.	Use WDI/BOP aggregate FDI and capital-flow proxies.
GAP-08	Labour and human capital	Sector/region employment, wages, vacancies, unemployment registry, education-employment linkage, skill indicators	Needed for labour models beyond aggregate employment/unemployment dashboards.	Use aggregate labour indicators and MoE social workbook.
GAP-09	Model governance	Official forecast vintages, model owner, approval history, and benchmark forecast archive	Needed to reproduce official tables, compare errors, and govern model changes inside MİİS.	Use available workbooks and public forecasts as benchmark evidence.
GAP-10	Structural models	Latest supply-use/input-output table, sectoral SAM, household/firm microdata, tax-benefit detail	Needed for credible CGE/DSGE/input-output models; otherwise only demonstrator possible.	Keep structural models as feasibility-stage demonstrators.

4 Current Ministry/CAEM Methodological Base

The supplied methodology is already organised around sector blocks: real sector, fiscal sector, social sector, monetary/prices and external sector. The forecast process uses scenario assumptions, sector equations and accounting identities. CAEM provides the financial-programming structure: macro consistency, sector accounts, fiscal/external/monetary linkages, trend-gap analysis and scenario simulation.

This means the recommended system should not replace the Ministry's existing model logic with an opaque black-box model. It should formalise, govern and extend it:

- **preserve identities** where the current methodology defines accounting relationships;
- **re-estimate transparent equations** where the Ministry already uses behavioural equations;
- **add benchmark models** to discipline forecast performance;
- **add shock models** for policy analysis under uncertainty;
- **gate advanced models** by diagnostics and data readiness.

5 Econometric and Analytical Framework

5.1 Statutory and Accounting Core

The first model layer is not statistical. It is the legally required accounting core. For output, the current methodology decomposes GDP by oil and non-oil blocks:

$$Y_t = Y_t^{oil} + Y_t^{non\text{-}oil} + T_t^{net}, \tag{1}$$

where Y_t is GDP, Y_t^{oil} is oil-gas value added, $Y_t^{non\text{-}oil}$ is non-oil value added, and T_t^{net} captures net product and import taxes where the classification requires it. Expenditure-side consistency is checked through:

$$Y_t = C_t + I_t + X_t - M_t. \tag{2}$$

The fiscal core uses:

$$BAL_t = R_t - E_t, \qquad D_t = (1 + i_t)D_{t-1} + E_t - R_t + A_t, \tag{3}$$

where BAL_t is the fiscal balance, R_t revenue, E_t expenditure, D_t debt, i_t the effective interest rate and A_t stock-flow adjustment. Production implementation requires official budget classification, treasury and debt data.

5.2 Reduced-Form Regression Library

The transparent econometric layer estimates equations of the form:

$$y_t = \alpha + \beta_1 x_{1t} + \cdots + \beta_k x_{kt} + \varepsilon_t, \tag{4}$$

with robust standard errors where appropriate. The purpose is not to maximise in-sample fit mechanically. The purpose is to quantify interpretable relationships and provide benchmark sensitivity estimates for GDP growth, inflation, external balance and budget indicators.

Table 3: Reduced-form regression library summary

Model	Dependent variable	N	R2	Adj. R2	Purpose
GDP growth equation	gdp_{}growth_{}pct	27	0.608	0.537	Reduced-form growth sensitivity to oil rents, inflation, exchange-rate pressure and external balance.
Inflation pass-through equation	inflation_{}cpi_{}pct	26	0.473	0.372	Inflation persistence and exchange-rate pass-through benchmark.
Current account equation	current_{}account_{}pct_{}gdp_{}wdi	27	0.361	0.277	External-balance sensitivity to oil rents and domestic activity.
Tax revenue equation	tax_{}revenue_{}pct_{}gdp	19	0.155	0.049	Public-data fiscal benchmark; not a replacement for tax-administration microdata.
Exports growth equation	exports_{}growth_{}pct	27	0.504	0.439	External nominal export-growth benchmark.

Table 4: Selected robust regression coefficients

Model	Term	Coef.	HC1 SE	t	p-value
GDP growth equation	oil_{}rents_{}pct_{}gdp	0.836	0.255	3.278	0.001
GDP growth equation	inflation_{}cpi_{}pct	-0.032	0.023	-1.345	0.179
GDP growth equation	fx_{}depreciation_{}pct	-0.069	0.104	-0.668	0.504
GDP growth equation	current_{}account_{}pct_{}gdp_{}wdi	-0.148	0.075	-1.962	0.05
Inflation pass-through equation	inflation_{}lag	0.042	0.004	11.19	0.0
Inflation pass-through equation	fx_{}depreciation_{}pct	0.177	0.089	1.984	0.047
Inflation pass-through equation	gdp_{}growth_{}pct	-0.04	0.136	-0.294	0.769

Model	Term	Coef.	HC1 SE	t	p-value
Inflation pass-through equation	oil_{}rents_{}pct_{}gdp	0.43	0.193	2.227	0.026

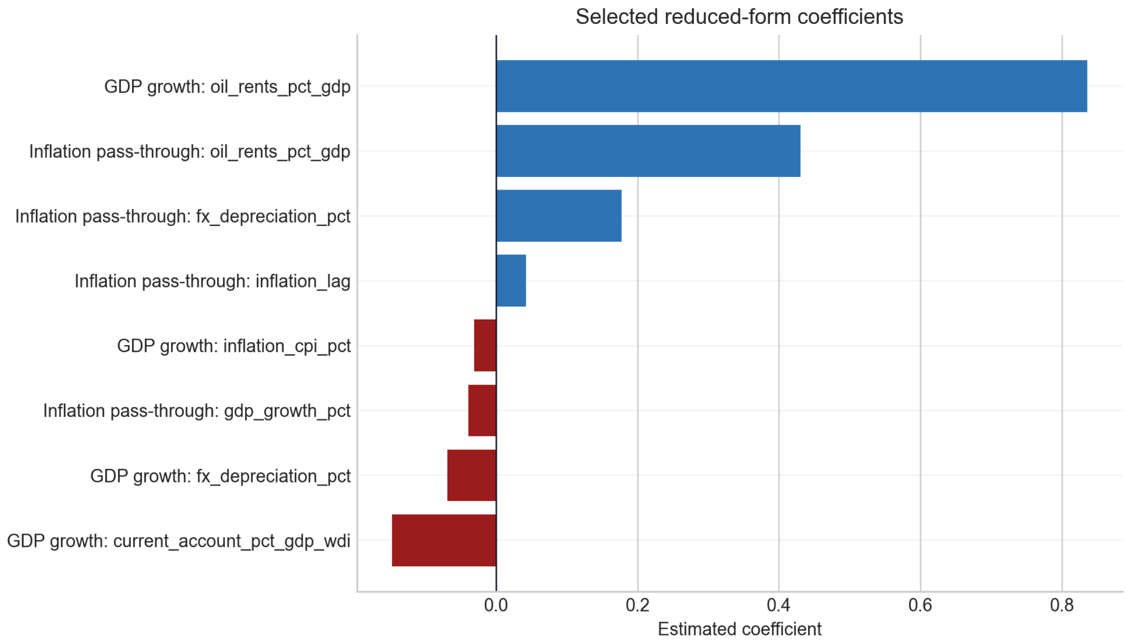

Figure 3: Selected reduced-form coefficients from the regression library. Coefficients must be interpreted as conditional empirical associations, not as structural causal multipliers unless identification is strengthened.

5.3 Recursive VAR and Impulse Responses

The shock-analysis layer uses a recursive VAR:

$$z_t = c + A_1 z_{t-1} + \cdots + A_p z_{t-p} + u_t, \qquad u_t = B\eta_t, \tag{5}$$

where z_t is the vector of quarterly variables, A_i are autoregressive matrices, u_t are reduced-form innovations and $B\eta_t$ is the recursive shock decomposition. The estimated ordering is Brent oil price, money, policy rate, CPI and non-extractive GDP proxy.

This should be described as a recursive short-run VAR with Cholesky identification. It should not be described as a fully structural SVAR unless the Ministry approves the structural restrictions and the restrictions are estimated and tested.

Table 5: Recursive VAR diagnostics

Start	End	Obs.	Lags	Stable
2002Q1	2024Q4	92	2	True

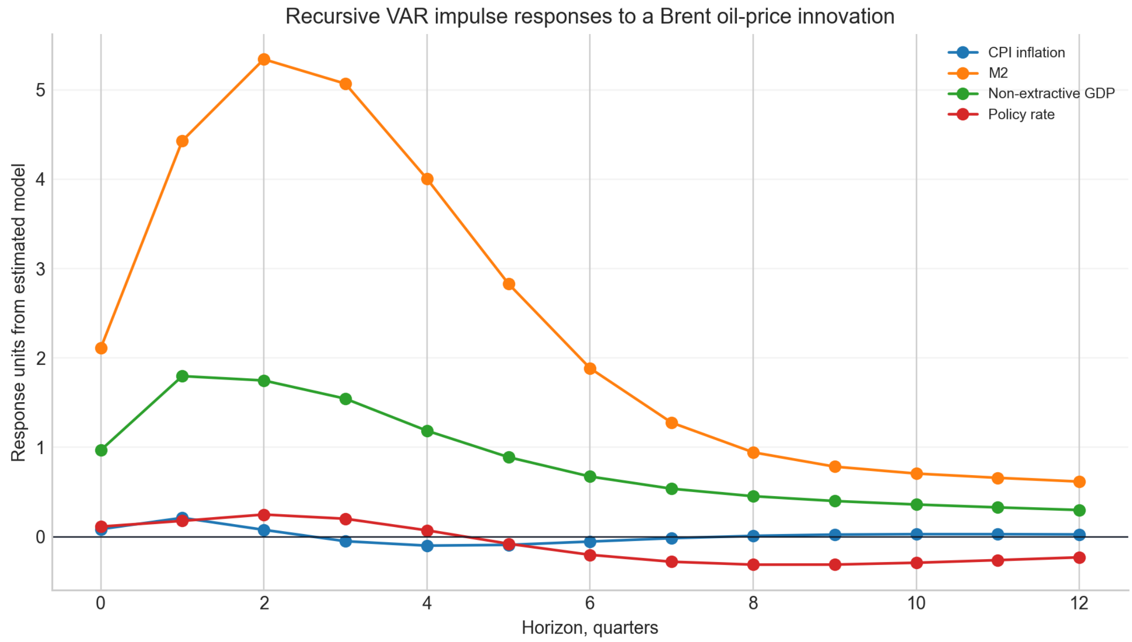

Figure 4: Impulse responses to a Brent oil-price innovation from the recursive VAR. Confidence bands should be added before the model is used for final policy quantification.

5.4 VECM and Cointegration Gate

A VECM has the form:

$$\Delta z_t = \Pi z_{t-1} + \Gamma_1 \Delta z_{t-1} + \cdots + \Gamma_{p-1} \Delta z_{t-p+1} + \varepsilon_t, \qquad \Pi = \alpha\beta', \tag{6}$$

where $\beta' z_{t-1}$ defines long-run cointegrating relationships and α measures adjustment to disequilibrium. VECM should be promoted only if the variables are integrated and the cointegration rank is empirically stable.

The current evidence is not yet sufficient for production promotion. The available Johansen output is sensitive to sample and variable choices, and one tested configuration rejects through the highest rank, which is more consistent with stationary behaviour/full-rank dynamics than with a clean low-rank cointegrating structure. Therefore the VECM should remain a research module until the Ministry supplies approved quarterly vintages and a final variable set.

Table 6: Johansen trace-test evidence

Rank	Trace statistic	95 percent critical	Reject at 95 percent
0	121.256	47.854	True
1	56.632	29.796	True
2	16.104	15.494	True
3	3.869	3.842	True

5.5 Machine Learning and Deep Learning Gate

Machine learning can be useful for short-horizon prediction, anomaly detection and risk scoring, but it should not be promoted merely because it is advanced. The promotion rule should be:

$$RMSE(m) < RMSE(b) \tag{7}$$

where m is the candidate model and b is a transparent benchmark such as a naive last-value forecast, ARIMA or a parsimonious regression. Interpretability and data governance are additional acceptance conditions.

The current benchmark result does not support promoting machine learning as the primary GDP forecast engine. The best model in the stored benchmark is Naive last value with RMSE 3.66, and the listed ML variants do not improve on it. ML should remain a benchmark/research layer until it repeatedly outperforms transparent models out of sample.

Table 7: Machine-learning forecast benchmark

Model	RMSE	MAE	Obs.
Naive last value	3.660	2.672	20
Ridge	4.645	3.930	20
SVR	4.645	3.741	20
OLS	4.660	3.376	20
Random Forest	5.150	4.444	20
LASSO	5.363	4.660	20
Gradient Boosting	5.689	4.400	20

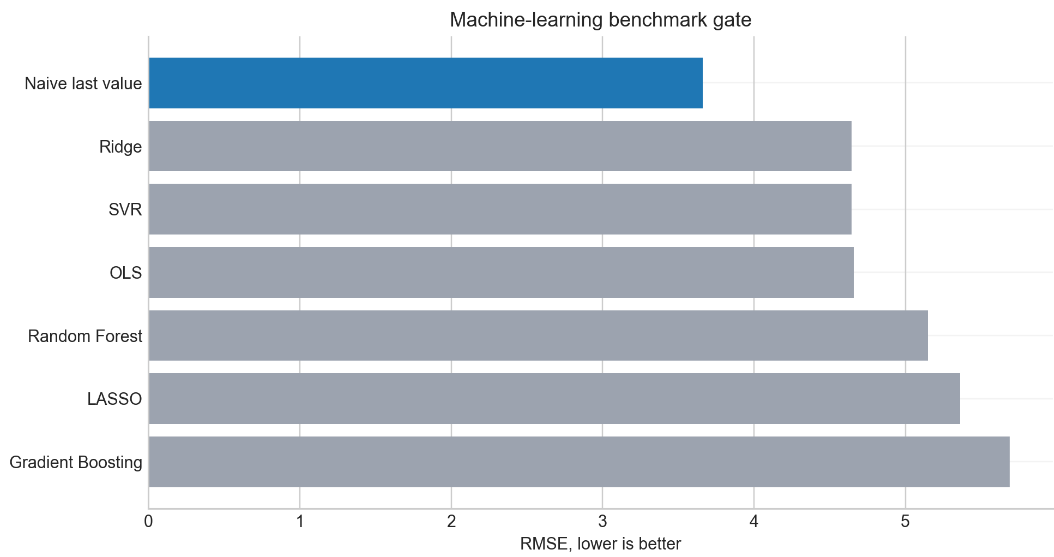

Figure 5: ML benchmark gate. In this result set, the simple naive benchmark is strongest, so ML is not promoted as the production forecast engine.

6 Scenario Analysis

The scenario engine must support the three statutory scenarios and additional shock scenarios requested in the advanced analytics material: oil-price shocks, exchange-rate shocks, fiscal shocks, external-demand shocks and tax-policy changes. Scenario assumptions must be stored, versioned and visible to the user.

For the pilot, the available scenario files already allow two layers:

- CAEM/statutory oil-price scenarios showing direct GDP and fiscal-oil-revenue effects;
- reduced-form macro scenarios showing GDP growth and inflation paths under alternative assumptions.

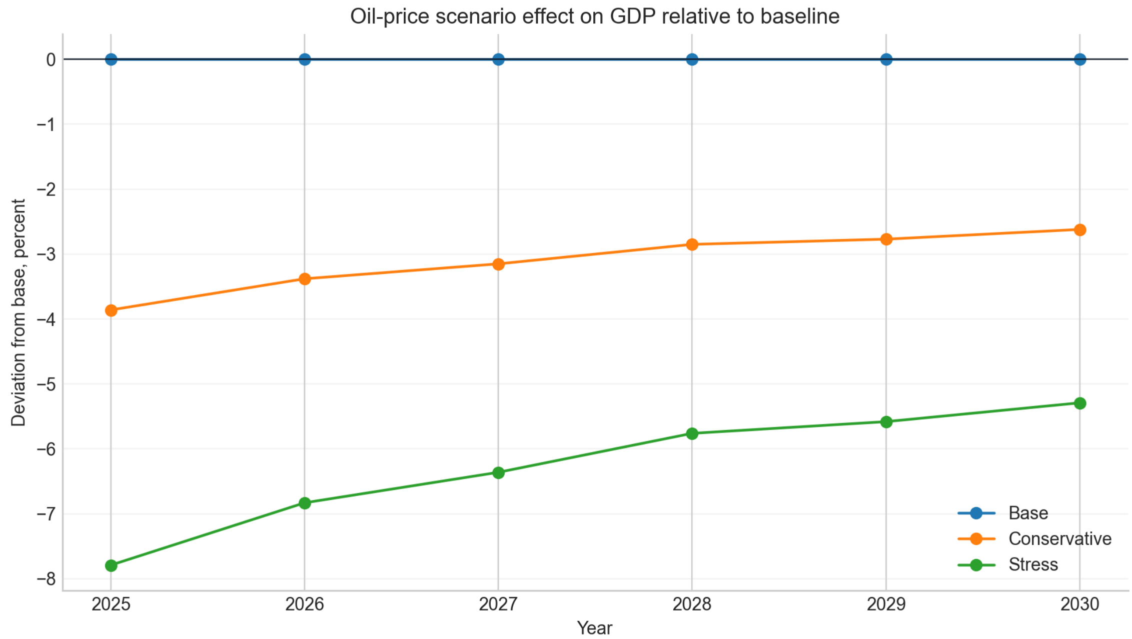

Figure 6: Oil-price scenario impact on GDP relative to the baseline. This is appropriate for the first pilot, provided assumptions are shown explicitly on the MİİS screen.

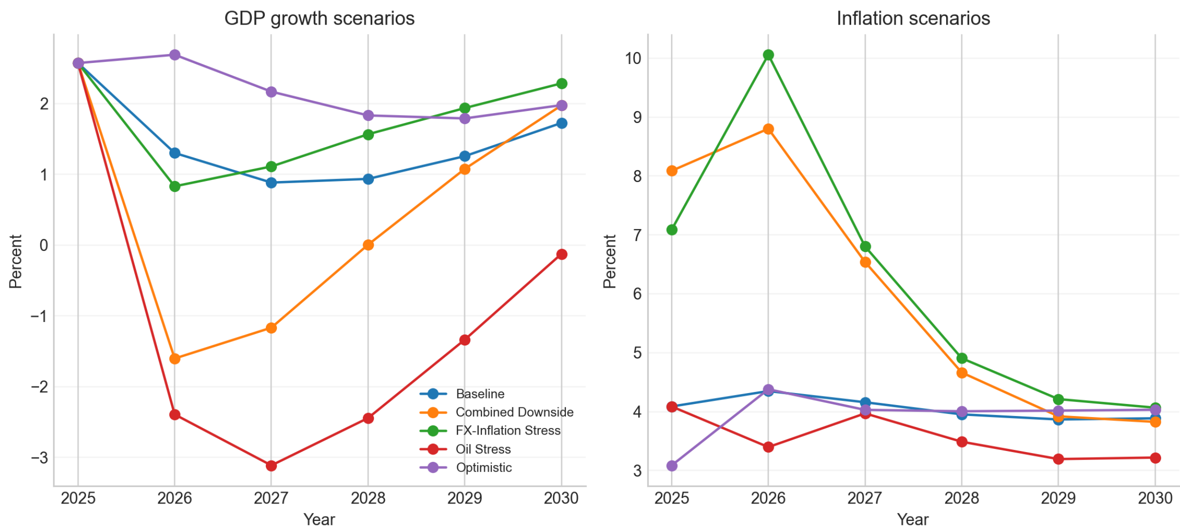

Figure 7: Scenario paths for GDP growth and inflation from the prior scenario engine. These are evidence for a working scenario layer, not official Ministry forecasts unless approved.

7 Databiz Model Traceability

The 13 Databiz model requests should be implemented as a traceable model catalogue rather than separate ungoverned equations. Each requested model needs a data mart, dependent variable, method, evidence source, data-gap status and pilot priority.

Table 8: Traceability matrix for the 13 Databiz model requests

No.	Block	Dependent variable	Recommended method	Implementation status	Priority
1	GDP	Nominal or real GDP	Start with accounting identity + transparent ARDL/ECM/OLS; add quarterly VAR shock layer after diagnostics.	Pilot feasible now with available annual/quarterly data; production requires official Ministry-approved vintages.	P1
2	GDP per capita	GDP per capita	Derived identity GDP/population plus scenario sensitivity; avoid overfitted causal model.	Feasible now as derived statutory indicator.	P1
3	Non-oil GDP	Non-oil GDP	Sectoral bridge/accounting model plus transparent regression; scenario layer for investment/export/fiscal shocks.	Pilot feasible; sector detail improves with official internal sector vintages.	P1
4	GDP growth	Real GDP growth rate	Benchmark suite: ARIMA/ARDL/VAR; promote model by backtest and diagnostics.	Pilot feasible; use 2026 as first forecast year.	P1
5	FDI	Foreign direct investment	Start with data mart + descriptive/benchmark model; causal model only after enough consistent data.	Partial pilot feasible; production needs richer institutional and incentive data.	P2
6	Capital flows	Net capital flows	External-sector accounting bridge and VAR-style shock analysis if frequency supports it.	Partial pilot feasible.	P2
7	Trade	Trade volume / turnover	Gravity-style/ARDL trade model with partner demand and exchange rate; benchmark against current MoE trade equations.	Pilot feasible; detailed production needs customs microdata.	P1
8	Exports	Exports, including non-oil exports	Separate oil and non-oil export equations; link oil exports to production/price assumptions.	Pilot feasible at aggregate level.	P1
9	Imports	Imports	Import demand equation plus accounting link to expenditure GDP.	Pilot feasible at aggregate level.	P1
10	Net trade / openness	Net exports and trade share of GDP	Derived identities from export/import/GDP; scenario charts.	Feasible now as derived indicator.	P1
11	Labour	Employment	Labour-demand equation linked to non-oil GDP and wages; sector bridge where data exist.	Pilot feasible with aggregate annual data.	P2
12	Labour	Unemployment	Okun-style benchmark plus demographic/labour-force accounting.	Pilot feasible at aggregate level.	P2
13	Labour / human capital	Labour-market and human-capital indicators	Descriptive dashboard first; causal models only after harmonised micro/sector data.	Partial pilot feasible.	P2

The technical prioritisation is clear. Models 1–4 and 7–10 are the first pilot because the e-mails explicitly prioritise GDP and foreign trade. Models 5–6 and 11–13 should be added after the data marts are governed and after the Ministry confirms whether sector/region/project-level data are available.

8 MİİS Implementation Architecture

The MİİS implementation should be built as a controlled system with seven layers:

1. **Source ingestion and validation:** load Ministry workbooks, CAEM, public data and internal administrative records.
2. **Versioned data marts:** GDP, trade, FDI/capital, labour, fiscal, scenario assumptions and model registry.
3. **Model engine:** re-estimation, diagnostics, benchmark comparison and forecast generation.
4. **Scenario engine:** baseline, pessimistic, optimistic and user-defined shocks.
5. **Legal output generator:** Decree-75 indicator tables and budget calendar outputs.
6. **Dashboard/reporting layer:** charts, tables, interpretation and downloadable outputs.

7. **Audit and approval workflow:** user, date, model version, data vintage and scenario ID for every forecast run.

Table 9: Proposed MİİS data marts

Data mart	Model block	Grain	Status
DM_{}GDP_{}CORE	GDP models 1-4	annual/quarterly indicator by vintage	proposed; partly populate now
DM_{}GDP_ {}DETERMINANTS	GDP models 1-4	period x determinant	proposed; partly populate now
DM_{}TRADE_ {}ANALYTICS	Trade models 7-10	period x flow x partner/product where available	proposed; aggregate pilot now, full production needs customs data
DM_{}EXPORT	Trade export model 8	period x export category	proposed; aggregate pilot now
DM_{}IMPORT	Trade import model 9	period x import category	proposed; aggregate pilot now
DM_{}FDI_{}CAPITAL_ {}FLOW_{}CORE	FDI/capital models 5-6	period x flow type x sector/origin where available	proposed; aggregate pilot only
DM_{}LABOUR_ {}MARKET_{}CORE	Labour models 11-13	period x labour indicator x sector/region where available	proposed; aggregate pilot now
DM_{}FISCAL_ {}BUDGET_{}CORE	Budget and macro-fiscal linkage	period x budget classification x scenario	proposed; prototype only until internal treasury/tax/debt data supplied
DM_{}SCENARIO_ {}ASSUMPTIONS	All scenario models	scenario x period x assumption	proposed; implement in pilot
DM_{}MODEL_ {}REGISTRY	Model governance	model version	proposed; implement in pilot

9 Model Governance

The governance decision is the most important protection against over-claiming. A model is promoted only if it is legally required, mechanically required by accounting identity, or empirically supported by diagnostics and backtesting.

Table 10: Model governance and promotion decisions

Model family	Decision	Reason	Caveat
Accounting/statutory macro identities	promote to pilot	Required for Decree-75 compliance and internally interpretable.	Must align all 2025 actuals and 2026 forecast start.
Transparent OLS/ARDL/ECM sector equations	promote conditionally	Matches current Ministry style and is explainable.	Use diagnostics, sample windows and out-of-sample benchmark before promotion.
ARIMA / benchmark forecasts	promote as benchmark	Useful baseline for forecast discipline.	Should not replace structural/sector knowledge where identities dominate.
Recursive VAR / Cholesky IRF	promote for scenario analysis with corrected wording	Useful for oil/FX/inflation shock narratives.	Call it recursive VAR unless true structural restrictions are imposed; add confidence bands.
VECM / cointegration	do not promote yet	Current Johansen evidence does not robustly support cointegration in the tested fiscal sample.	Re-test with approved quarterly/annual series before using VECM.
Machine learning models	research/pilot benchmark only	Some models beat simple benchmarks in limited tests, but policy use needs stability and interpretation.	Promote only if repeat backtests outperform transparent models.
Deep learning / LSTM / Prophet / TFT	not promoted for pilot	Stored results show simple benchmarks outperform deep-learning variants in current sample.	Can remain as optional research after data expansion.
Bayesian / state-space	promote as uncertainty/nowcast research component	Useful for uncertainty and filtering if documented carefully.	Needs official quarterly vintages and validation.
Input-output / CGE / DSGE	feasibility-stage only	Requested in advanced analytics concept but not supportable as production without structural tables/microdata.	Label all current outputs as calibrated demonstrators.

Model family	Decision	Reason	Caveat
Budget execution engine	prototype only until internal data supplied	Logic can be designed, but production requires tax/customs/treasury/debt/fund data.	Do not claim full budget-execution model from public aggregates.

The resulting policy is:

- statutory/accounting identities are mandatory;
- transparent OLS/ARDL/ECM models are promoted conditionally;
- ARIMA and naive models are retained as benchmarks;
- recursive VAR is promoted for scenario interpretation with careful wording;
- VECM is not promoted until cointegration evidence is stable;
- ML and deep learning remain research/benchmark layers unless they beat transparent baselines;
- CGE, DSGE and input-output models require structural tables and calibration data before production use;
- the budget execution engine is prototype-only until internal administrative data are supplied.

10 Implementation Roadmap and Acceptance Tests

The roadmap is designed to be compatible with the Ministry request and with the practical MİİS implementation sequence. It starts with governance and pilot data marts, then model and scenario functionality, then budget linkage and advanced analytics.

Table 11: Recommended implementation roadmap

Phase	Window	Objective	Deliverable	Exit gate
P0	Immediate / 1 June response	Agree scope, model list, data gaps and pilot priorities	technical response memo + matrices	client confirms scope and data owners
P1	Weeks 1-4	Implement GDP and trade data marts plus model registry	GDP/trade pilot in MİİS design	data validation passes and model equations reproducible
P2	Weeks 5-10	Add scenario engine, diagnostics and benchmark model selection	forecast/scenario workspace	baseline and scenarios reconcile with CAEM/statutory indicators
P3	Weeks 8-16	Add fiscal/budget linkage where data are supplied	budget prototype moving toward production module	tax/customs/treasury/debt data available and legal outputs match templates
P4	After core pilot approval	Advanced analytics extension	ML/risk/structural modules only where validated	advanced model outperforms or serves distinct accepted policy need

Table 12: Acceptance tests for the MİİS pilot

Phase	Gate	Acceptance test
P0	Scope and data governance	signed model scope, source registry, data owner/vintage fields defined
P1	GDP and trade pilot data marts	requested GDP/trade variables either populated or explicitly marked missing
P1	Model registry	all 13 Databiz models mapped to method, data, status and gap
P1	Forecast start vintage	2025 treated as actual where official actuals exist; 2026 is first forecast year
P2	Scenario engine	baseline and shock assumptions visible; outputs reconcile with CAEM/accounting identities
P2	Econometric promotion gate	model beats benchmark or is required by accounting/legal identity; otherwise marked research/not promoted

Phase	Gate	Acceptance test
P3	Budget module	macro-to-budget prototype works on aggregate data; production waits for tax/customs/treasury/debt data
P3	Advanced analytics	AI/ML/DSGE/CGE/IO modules labelled pilot/research unless validated on required data

11 Final Recommendations

1. **Approve GDP and trade as the first pilot.** These blocks are specifically requested, data-supported and foundational for budget and scenario analysis.
2. **Set the forecast vintage rule.** Treat available 2025 official actuals as actuals and start forecasts in 2026.
3. **Create the data mart and model registry before adding more models.** This prevents the system from becoming a set of disconnected spreadsheets.
4. **Adopt an EViews-like functional benchmark, not necessarily EViews as the platform.** MİİS should support workfiles/data marts, equations, re-estimation, diagnostics, scenarios, forecast storage and audit.
5. **Use advanced analytics selectively.** ML, VECM, DSGE, CGE and AI should be accepted only where the evidence and data support them.
6. **Request internal data immediately.** The production budget module requires tax, customs, treasury, debt, SOFAZ/off-budget fund and project-level investment data.

Source Note

This report is built from the controlled local source package: Ministry-supplied CAEM workbook and documentation, Decree No. 75 and budget-rule references, Ministry methodology documents, Databiz model notes, prior model-output CSV files, public macro/fiscal datasets already collected in the project folder, and the executed notebook package. The generated figures and tables are produced programmatically by `scripts/build_assets.py`. The LaTeX source and assets are kept together so the PDF can be regenerated after any data update.